Extinction Archive

Umwelt memorial for the woolly mammoth

Biodesign Challenge 2026 · Biodigital Excellence · *Convergent Life*

Digital Art & AI Technology (MDes) · 2-person team · 2–3 week MVP

What we lose when a species disappears

- Not only **bodies**, but **ways of living in time**
- Seasonal routes · light–dark entrainment · social rhythms · **sensory worlds (*Umwelt*)**
- Science stays in papers; public memory flattens into icons

Design question: How do we make rigorous biology **feelable** without lying?

Project thesis

Extinction Archive = browser-first experience that builds a **hypothesis-grade** *Umwelt* for ***Mammuthus primigenius***

- **Grounded** in citable research (tusk isotope stratigraphy, Arctic adaptation, extinction context)
- **Expressed** through shaders, controlled generative passes, and **Web Audio**
- **Closed** with **evidence literacy** + **branching ethics** — digital recall must not erase conservation urgency

Why the woolly mammoth (locked scope)

Pillar	Why it fits the MVP
Isotope “diaries”	Published life-history reconstructions → **Dossier foldout + R/ggplot stills**
Photoperiod / season	Arctic extremes → **lighting + Mixer pulse**
Biodigital story	Paleogenomics / morphology literature ↔️ **parametric scene + labels**
Ethics	De-extinction & Arctic narratives → **sharp, honest branch cards**

Frozen: single hero species for 2–3 weeks · no parallel dossiers

User journey (three modules)

```
Dossier  →  Mixer  →  Ethics
 (read)     (listen)   (choose)
```

Group1Slim (locked): all three ship · Ethics ~**2 min** in live talk

Module A — *Dossier*

- **Anchors:** geography & time (LGM–Early Holocene framing — specific sites **Cited**)
- **Science foldout:** Sr/O (etc.) tusk chemistry · published individual tracks
- **Labels on every block: Cited / Modeled / Speculative**
- **360° scene** driven by **season & sun parameters** — not unbounded “text-to-world”

Module B — *Mixer*

- **≥3 audio layers (mammoth-tuned):**
  - Seasonal / **photoperiod pulse**
  - Steppe–tundra **environment bed** (*Speculative* where elephant-relative)
  - **Anthropocene pressure** layer (*Cited/Modeled* where possible)
- **Harmony → conflict:** simple **DSP rules** (e.g. gain competition → ducking / beating)
- **Documented:** rules on a slide + README (judges can interrogate the “black box”)

Module C — *Ethics console*

C gate (~45 s): three true/false or sort prompts — *"Is this claim Cited or Speculative?"*

B branches (~75 s): 2–3 cards, e.g.

- Conservation funding vs de-extinction hype
- Invasive-species / ecosystem management frame
- Data, land, Indigenous knowledge — **questions + team commitments** (no speaking for others)

Outcome: different **finale mix preset** + short caption (≤80 words)

Bio + digital — both load-bearing

Biology (from literature)	Digital (implementation)
Isotope profiles, seasonal life	`ggplot` + Dossier charts
Light environment, habitat proxies	Three.js / R3F + shaders
Uncertainty & debate	**C/M/S** UI + copy
Emotional pacing	Mixer + branch endings

Remove biology → fancy demo · **Remove code** → static essay → **both fail** → we pass the biodigital test

Physical + AR (lightweight)

- **3D print:** abstract **tusk “growth core”** or skull token — material witness
- **Mobile WebAR:** same GLB + stems as desktop (`?ar=1` or entry route)
- **Fallback:** QR → same Dossier anchor (MoMA-grade reliability)

Tech stack (suggested)

- **Front:** Vite · React · React Three Fiber / Three.js · Web Audio API
- **Science viz:** R · ggplot2 → exported PNG/SVG for deck + site
- **Gen AI: Control-first** passes (masks/palette) + **procedural base** · API costs checked
- **AR:** WebXR-friendly path (A-Frame acceptable if team-standard)

2–3 week schedule (summary)

Week	Focus
1	Stack · Dossier UI + mammoth JSON · scene · Mixer prototype · evidence audit #1 · ggplot v1
2	Ethics UI · GLB · WebAR minimal · print · Methods/Cite/Limits slides · rehearsal · trailer
3*	Polish · a11y · mock Q&A · P0 bugs only · *optional stretch*

Compress weeks 2–3 if only 14 days.

BDC rubric — how we aim to score

Dimension	Our hook
Narrative	Hero specimen arc · live demo · honest labels
Concept	*Umwelt* + isotope-time · biodigital integration
Context	De-extinction ethics · moral hazard of “reversible” extinction · conservation framing
Reflection	Documented iterations · Speculative gates · limits slide

Target: **Biodigital Excellence** prize narrative

Deliverables checklist (course / BDC-oriented)

- [ ] **Talk** · 8 min + 2 min demo
- [ ] **Visuals** · UI capture + stills
- [ ] **Physical** · print + AR footage
- [ ] **Trailer** · 45–90 s
- [ ] **Deck** · includes **Methods · Citations · Limitations**

Methods

- **Evidence base:** three peer-reviewed anchors — **tusk isotope life history** (Alaska bull mammoth; Sr/O maps + random-walk model); **comparative genomes** (mammoth vs elephant circadian & thermal pathways); **extinction drivers** (climate × human pressure modeling)
- **Dossier:** claims imported to JSON; every block tagged **Cited / Modeled / Speculative**; `ggplot2` reproduces **public tusk isotope / time** summaries for stills (not a re-identification of the individual)
- **3D scene:** sun height, fog, palette driven by **season & photoperiod** parameters; optional **control-net / masked** image pass logged with **model ID + date + seed**
- **Mixer:** layered stems; DSP rules documented (release **v0.1** in README); elephant-relative beds labeled *Speculative*
- **Process:** ≥2× **30 min evidence audits** (Week 1 & 2) · GEN-AI & prompt changelog for Reflection slide

Citations — core papers (3) + tooling (+1)

1 — Tusk isotope mobility (Dossier science fold · migration model)

Wooller, M. J., Bataille, C., Druckenmiller, P., Erickson, G. M., Groves, P., Haubenstock, N., Howe, T., Irrgeher, J., Mann, D., Moon, K., Potter, B. A., Prohaska, T., Rasic, J., Reuther, J., Shapiro, B., Spaleta, K. J., & Willis, A. D. (2021). Lifetime mobility of an Arctic woolly mammoth. *Science*, *373*(6556), 806–808. https://doi.org/10.1126/science.abg1134

2 — Genomes & Arctic adaptation (circadian / thermal framing)

Lynch, V. J., Bedoya-Reina, O. C., Ratan, A., Sulak, M., Drautz-Moses, D. I., Perry, G. H., Miller, W., & Schuster, S. C. (2015). Elephantid genomes reveal the molecular bases of Woolly Mammoth adaptations to the Arctic. *Cell Reports*, *12*(2), 217–228. https://doi.org/10.1016/j.celrep.2015.06.027

3 — Extinction context (climate × humans; avoid single-cause)

Nogués-Bravo, D., Rodríguez, J., Hortal, J., Batra, P., & Araújo, M. B. (2008). Climate change, humans, and the extinction of the Woolly Mammoth. *PLoS Biology*, *6*(4), e79.

Limitations (honest — judges reward this)

- No living mammoth — **all sensory reconstruction is partial**; elephant-relative regions are **Modeled/Speculative**
- Generative media can **feel real** — we **design against** deceptive realism
- **2–3 weeks / 2 people** → one species, narrow but deep
- AR hardware variance — **desktop path is canonical**

Team & roles (flat + audits)

- **Cross-functional** both members · daily **15 min** sync
- **Evidence audits** 2× weekly (30 min) — no unlabeled claims in demo
- Optional split: **viz/code** vs **sound/research/copy** — but both own the rubric

Thank you / Q&A prompts we practice

- Why mammoth and not a “easier” icon species?
- Where exactly is the **living biology**? (Answer: **from biology** evidence + optional stretch plant; thesis is digital–paleo bridge.)
- How do you prevent **misinformation**? (**C/M/S** + ethics gate + limitations)
- What would you do with **4 more months**?

Backup — one-line pitch

> **Extinction Archive** turns tusk-time and Arctic light into a **listenable, clickable memorial** — and forces you to decide what “bringing the mammoth back” should cost.

Deck exports

English sources: `slides/BDC_Extinction_Archive_Mammoth.md` · **Chinese:** `slides/BDC_Extinction_Archive_Mammoth_zh.md`

Built files: `slides/export/BDC_Deck_EN.pdf` · `BDC_Deck_ZH.pdf` · `BDC_Deck_EN.pptx` · `BDC_Deck_ZH.pptx`

Preview hub: open `slides/preview.html` in a browser (see `slides/README.md`).